This assignment will not be graded and is only for practice.

1. Key Points:

Calculating minors and cofactors of a matrix:
If A is a square matrix, then the minor of the entry in the i-th row and j-th column (denoted by M_{ij}) is the determinant of the submatrix formed by deleting the i-th row and j-th column.
The ij-th cofactor (denoted by C_{ij}) is defined to be $(-1)^{i+j}M_{ij}$.
Calculating the determinant:

$det(A) = \sum_{i=1}^{n} a_{ij}(-1)^{i+j}M_{ij} = \sum_{i=1}^{n} a_{ij}C_{ij}$
Let us consider a 3×3 matrix A as follows:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Calculating M_{11} and C_{11}:

• Find out the sub-matrix by deleting the first row and the first column:
$$\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & a_{22} & a_{23} \\ \cdot & a_{32} & a_{33} \end{bmatrix}$$

• Calculating the determinant of the sub-matrix:
$$det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}\right) = a_{22}a_{33} - a_{32}a_{23}$$
$M_{11} = a_{22}a_{33} - a_{32}a_{23}$
$C_{11} = (-1)^{1+1}(a_{22}a_{33} - a_{32}a_{23}) = a_{22}a_{33} - a_{32}a_{23}$
Calculating M_{23} and C_{23} :

• Find out the sub-matrix by deleting the second row and the third column:
$$\begin{bmatrix} a_{11} & a_{12} & \cdot \\ \cdot & \cdot & \cdot \\ a_{31} & a_{32} & \cdot \end{bmatrix}$$
• Calculating the determinant of the sub-matrix:
$$det\left(\begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{bmatrix}\right) = a_{11}a_{32} - a_{31}a_{12}$$
$M_{23} = a_{11}a_{32} - a_{31}a_{12}$
$C_{23} = (-1)^{2+3}(a_{11}a_{32} - a_{31}a_{12}) = -a_{11}a_{32} + a_{31}a_{12}$
Calculating the determinant of A:

Expanding with respect to the first row:

$$det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$$

Expanding with respect to the second row:

$$det(A) = a_{21}(-1)^{2+1}M_{21} + a_{22}(-1)^{2+2}M_{22} + a_{23}(-1)^{2+3}M_{23}$$

Expanding with respect to the third row:

$$det(A) = a_{31}(-1)^{3+1}M_{31} + a_{32}(-1)^{3+2}M_{32} + a_{33}(-1)^{3+3}M_{33}$$

The determinant can also be calculated by expanding along columns.

Expanding with respect to the first column:

$$det(A) = a_{11}(-1)^{1+1}M_{11} + a_{21}(-1)^{2+1}M_{21} + a_{31}(-1)^{3+1}M_{31}$$

1) Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$. Find out M_{23}.

[Hint: Find out the sub-matrix by deleting the second row and the third column: $\begin{bmatrix} -1 & 2 & \cdot \\ \cdot & \cdot & \cdot \\ -1 & 0 & \cdot \end{bmatrix}$ Calculate the determinant of the sub-matrix.]

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

2) Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$. Find out C_{32}.

[Hint: Find out the sub-matrix by deleting the third row and the second column: $\begin{bmatrix} -1 & \cdot & 0 \\ 2 & \cdot & -1 \\ \cdot & \cdot & \cdot \end{bmatrix}$

Calculate the determinant of the sub-matrix to find M_{32}. To find the cofactor C_{23} the sign has to be taken care of as follows: $C_{32} = (-1)^{3+2}M_{32}$]

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

3) Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$. Find out the determinant of A.

[Hint: Expanding with respect to the first row:

$$det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$$

Expanding with respect to any row will also give you the answer]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -2

1 point

4) Consider the following square matrices:

1 point

$$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$$

Choose the set of correct options.

Hint:

B can be obtained by interchanging the first two rows.

C can be obtained by multiplying 2 to the first row of B.

- [ ] $det(B) = det(A)$.
- [ ] $det(A) = -det(B)$.
- [ ] $det(C) = -2det(B)$.
- [ ] $det(C) = -2det(A)$

No, the answer is incorrect.

Score: 0

Feedback:

• Sign of the determinant changes due to interchange of two rows.

• If a scalar c is multiplied with a row of a matrix, then the determinant of the new matrix will be c times the determinant of the earlier matrix.

Accepted Answers:

$det(A) = -det(B)$.
$det(C) = -2det(B)$.
$det(C) = -2det(A)$

2. Key Points:

Cramers' Rule: (This rule is applied to any system of linear equations with n equations and n variables. Here we recall the method for a system of linear equations with 3 equations and 3 variables.)

Consider a system of linear equations as follows: $$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

Let A_{x_i} be the matrix obtained by replacing the i-th column of A $(i.e., \begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix})$ by b, for $i = 1, 2, 3$.

$$A_{x_1} = \begin{bmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{bmatrix}$$
$$A_{x_2} = \begin{bmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{bmatrix}$$
$$A_{x_3} = \begin{bmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{bmatrix}$$

If $det(A) = 0$, then the solutions to the above system are $x_i = \dfrac{det\ A_{x_i}}{det\ A}$, for $i = 1, 2, 3$. i.e., $x_1 = \dfrac{det\ A_{x_1}}{det\ A}$, $x_2 = \dfrac{det\ A_{x_2}}{det\ A}$, $x_3 = \dfrac{det\ A_{x_3}}{det\ A}$

Consider a system of linear equations $$\begin{aligned} x_1 + x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 1 \\ -x_2 + x_3 &= 1 \end{aligned}$$

Let matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Let A_{x_i} be the matrix obtained by replacing the i-th column of A (i.e., $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$) by b, for $i = 1, 2, 3$.

Use the above information to answer the following questions(5,6):

5) Choose the set of correct options **_1 point_**

- [ ] $A_{x_1} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$
- [ ] $A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$
- [ ] $A_{x_2} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$
- [ ] $A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

6) Choose the set of correct options about the solutions to this system. **_1 point_**

- [ ] $x_1 = -2.$
- [ ] $x_2 = -2.$

- [ ] $x_3 = 3$.
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x_1 = -2$.
$x_3 = 3$.

3. Key Points:

Calculating the inverse of a matrix A:

Find out the cofactor matrix C, whose ij-th element is C_{ij}: the ij-th cofactor of A.
Adjoint of A is transpose of C.
Calculate the determinant of A and check whether it is non-zero or not.
If determinant is non-zero then the inverse of A exists and is given by

$$A^{-1} = \frac{1}{det(A)} adj(A).$$

7) Which of the following is the cofactor matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$? **1 point**

[Hint: Recall the steps to calculate the cofactors, given in Key Points 1.]

- ○ $\begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & -1 \\ -2 & 0 & 2 \end{bmatrix}$
- ○ $\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$
- ○ $\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$

- [ ] $\begin{bmatrix} 1 & -1 & -1 \\ -1 & 3 & -1 \\ -2 & 0 & 2 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$

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8) Which of the following is the adjoint matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$? ***1 point***

[Hint: Find the transpose to the cofactor matrix.]

- [ ] $\begin{bmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix}$
- [ ] $\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$
- [ ] $\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$
- [ ] $\begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$

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9)

Find out the determinant of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$.

[Hint: Recall the steps to calculate the determinant, given in Key Point 1.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

10) Which of the following is the inverse matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$? ***1 point***

[Hint: If determinant is non-zero then the inverse of A exists and is given by $A^{-1} = \frac{1}{det(A)} adj(A)$.]

- $\frac{1}{2}\begin{bmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix}$
- $\frac{1}{2}\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$
- $\frac{1}{2}\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$
- $\frac{1}{2}\begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{1}{2}\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$

11) Choose the set of correct options. **1 point**

Hint:
$A^{-1} = \frac{1}{det(A)} adj(A)$.
$det(A^{-1}) = \frac{1}{det(A)}$

- [ ] If $A = A^{-1}$, then $det(A)$ must be 1.
- [ ] If $A = A^{-1}$, then A must be the identity matrix.
- [ ] If $A = A^{-1}$, then $A^2 = I$.
- [ ] If $A^{-1} = adj(A)$, then $det(A)$ must be 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $A = A^{-1}$, then $A^2 = I$.
If $A^{-1} = adj(A)$, then $det(A)$ must be 1.

Your score is: 0/11